member of one group that enters another patch and mates with the local population there. Isolation for too long pushes local populations apart to the point where they can no longer mate with one another, and in time they become separate species or subspecies.

In the 1990s, Finnish ecologist Ilkka Hanski showed that at the core of metapopulation theory is the notion that local populations are unstable. The metapopulation as a whole may well be stable, but the local populations are likely to rise and fall in their individual patches in response to inside and outside influences. Some patch members may emigrate and join a much reduced population in danger of extinction, giving it renewed strength—a metapopulation feature known as the “rescue effect.” Other groups may completely vanish, leaving vacant patches for another population to recolonize. Hanski argued that there is persistent balance between “deaths” (local extinctions) and “births” (the establishment of new populations at unoccupied sites). He likened this balance to the spread of disease, with the susceptible and the infected representing in turn empty and occupied “patches” for disease-carrying parasites.

Ecologists see the concept of metapopulations as increasingly important in understanding how species will survive, particularly in the face of human influence on habitats. The theory helps them analyze the way populations rise and fall, using mathematical models to play out interactions, and enables them to predict how much habitat fragmentation a species can endure before it is driven to extinction.